

CompTIA.

A+ Core 2

Exam 220-1202

A+ 220-1202

Module 14



Supporting Windows

Prepare for A+ Core 2 by:

- Performing OS installations and upgrades
- Installing and configuring applications
- Troubleshooting Windows OS problems

Lesson 14.1

Troubleshoot Windows Networking

Troubleshoot IP Configuration (Slide 1 of 2)

Limited IP connectivity

- Verify IP config
- DHCP Settings and server availability (APIPA)

No Internet access

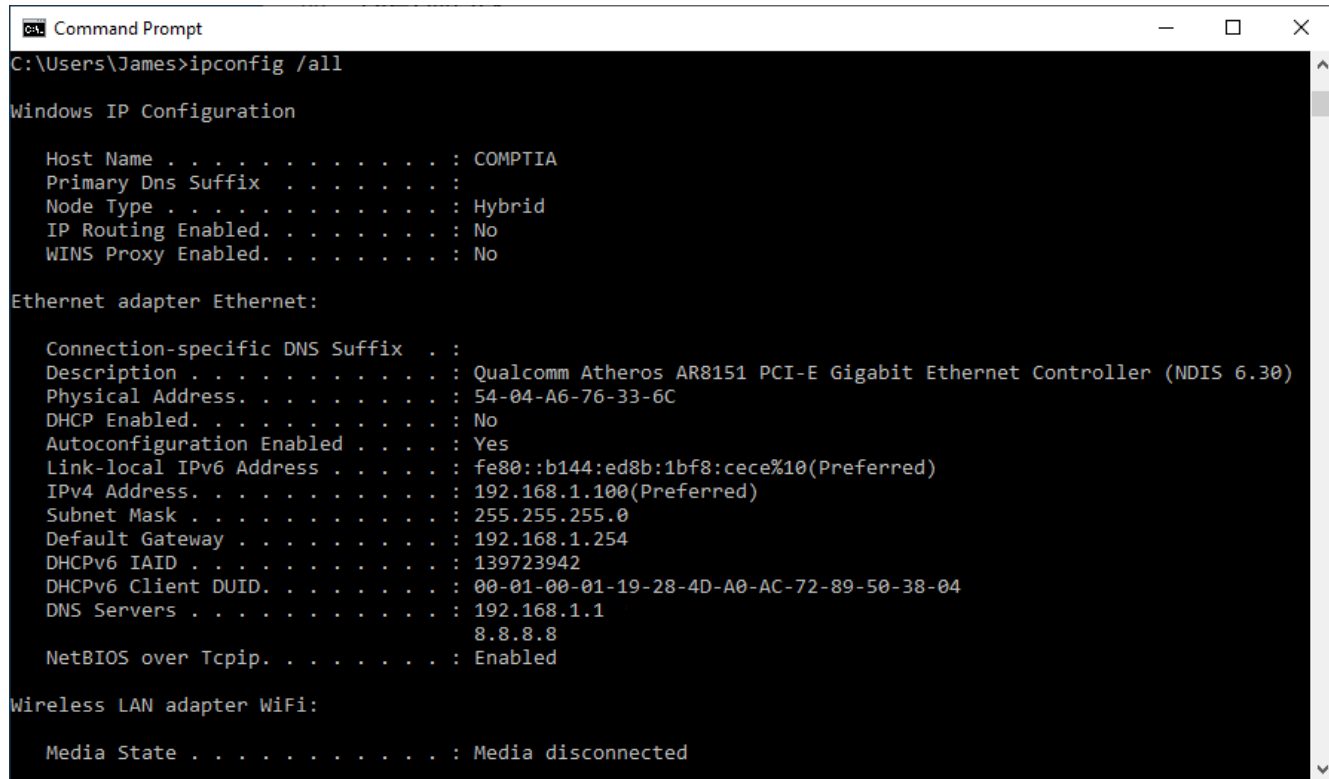
- IP configuration
 - IP/Subnet Mask/Gateway
- DNS server configuration

ipconfig command

- /all, /release, /renew, /displaydns, and /flushdns

hostname command

Troubleshoot IP Configuration (Slide 2 of 2)



```
Command Prompt
C:\Users\James>ipconfig /all

Windows IP Configuration

    Host Name . . . . . : COMPTIA
    Primary Dns Suffix . . . . . :
    Node Type . . . . . : Hybrid
    IP Routing Enabled. . . . . : No
    WINS Proxy Enabled. . . . . : No

Ethernet adapter Ethernet:

    Connection-specific DNS Suffix  . :
    Description . . . . . : Qualcomm Atheros AR8151 PCI-E Gigabit Ethernet Controller (NDIS 6.30)
    Physical Address. . . . . : 54-04-A6-76-33-6C
    DHCP Enabled. . . . . : No
    Autoconfiguration Enabled . . . . : Yes
    Link-local IPv6 Address . . . . . : fe80::b144:ed8b:1bf8:cece%10(Preferred)
    IPv4 Address. . . . . : 192.168.1.100(Preferred)
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 192.168.1.254
    DHCPv6 IAID . . . . . : 139723942
    DHCPv6 Client DUID. . . . . : 00-01-00-01-19-28-4D-A0-AC-72-89-50-38-04
    DNS Servers . . . . . : 192.168.1.1
                           8.8.8.8
    NetBIOS over Tcpip. . . . . : Enabled

Wireless LAN adapter WiFi:

    Media State . . . . . : Media disconnected
```

Screenshot courtesy of Microsoft.

Troubleshoot Local Network Connectivity (Slide 1 of 3)

Connectivity test

- Ping command
- Loopback IP, local workstation IP, then gateway IP, then remote system FQDN or IP

Local loopback or local workstation IP unreachable

- Troubleshoot network adapter configuration

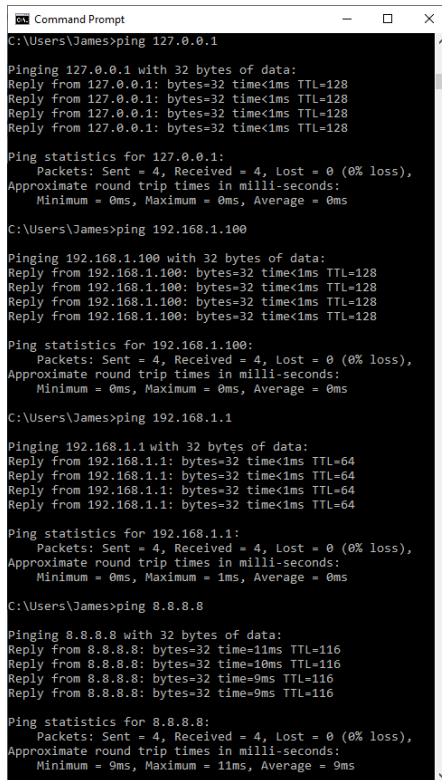
Gateway IP unreachable

- DHCP server options configured correctly
- Router IP configuration done correctly

Remote system FQDN or IP unreachable

- Verify resource exists
- Verify DNS settings

Troubleshoot Local Network Connectivity (Slide 2 of 3)



```
Command Prompt
C:\Users\James>ping 127.0.0.1

Pinging 127.0.0.1 with 32 bytes of data:
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128

Ping statistics for 127.0.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\Users\James>ping 192.168.1.100

Pinging 192.168.1.100 with 32 bytes of data:
Reply from 192.168.1.100: bytes=32 time<1ms TTL=128
Reply from 192.168.1.100: bytes=32 time<1ms TTL=128
Reply from 192.168.1.100: bytes=32 time<1ms TTL=128
Reply from 192.168.1.100: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.100:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\Users\James>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:
Reply from 192.168.1.1: bytes=32 time<1ms TTL=64
Reply from 192.168.1.1: bytes=32 time<1ms TTL=64
Reply from 192.168.1.1: bytes=32 time<1ms TTL=64
Reply from 192.168.1.1: bytes=32 time<1ms TTL=64

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

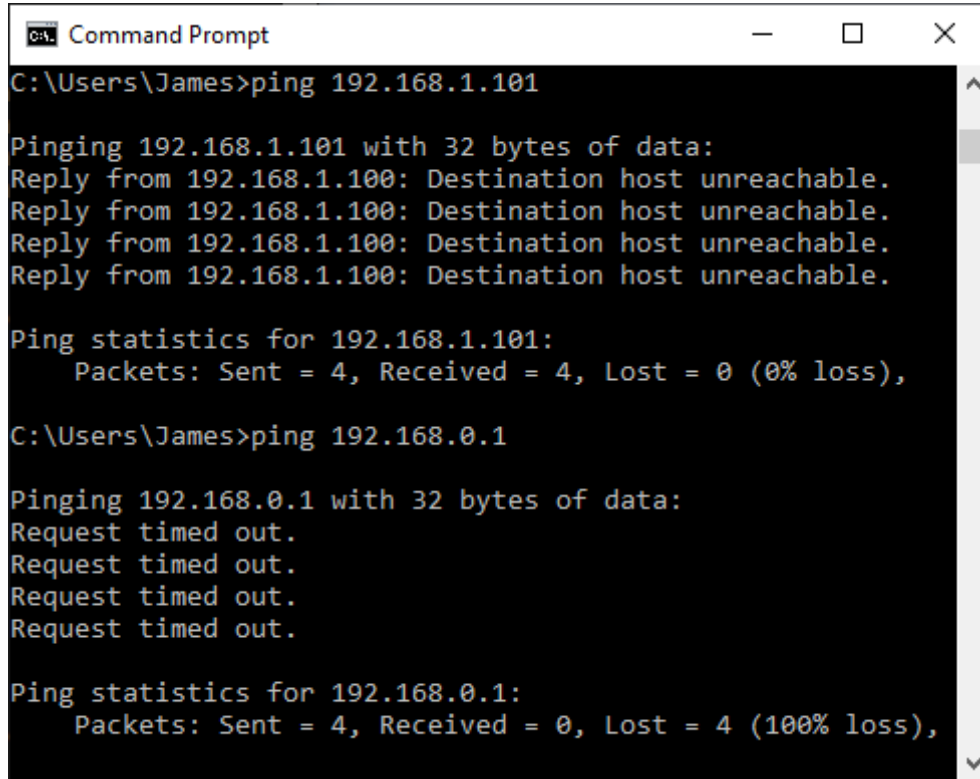
C:\Users\James>ping 8.8.8.8

Pinging 8.8.8.8 with 32 bytes of data:
Reply from 8.8.8.8: bytes=32 time=11ms TTL=116
Reply from 8.8.8.8: bytes=32 time=10ms TTL=116
Reply from 8.8.8.8: bytes=32 time=9ms TTL=116
Reply from 8.8.8.8: bytes=32 time=9ms TTL=116

Ping statistics for 8.8.8.8:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 9ms, Maximum = 11ms, Average = 9ms
```

Screenshot courtesy of Microsoft.

Troubleshoot Local Network Connectivity (Slide 3 of 3)



```
Command Prompt

C:\Users\James>ping 192.168.1.101

Pinging 192.168.1.101 with 32 bytes of data:
Reply from 192.168.1.100: Destination host unreachable.
Reply from 192.168.1.100: Destination host unreachable.
Reply from 192.168.1.100: Destination host unreachable.
Reply from 192.168.1.100: Destination host unreachable.

Ping statistics for 192.168.1.101:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

C:\Users\James>ping 192.168.0.1

Pinging 192.168.0.1 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.0.1:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

Screenshot courtesy of Microsoft.

Troubleshoot Remote Network Connectivity

```
C:\Users\James>tracert 192.168.1.1

Tracing route to eeHub.home [192.168.1.1]
over a maximum of 30 hops:

  1      1 ms      <1 ms      <1 ms  eeHub.home [192.168.1.1]

Trace complete.

C:\Users\James>tracert 8.8.8.8

Tracing route to dns.google [8.8.8.8]
over a maximum of 30 hops:

  1      <1 ms      <1 ms      <1 ms  eeHub.home [192.168.1.1]
  2      4 ms       5 ms       4 ms   172.16.16.15
  3      *          *          *      Request timed out.
  4      9 ms       9 ms       9 ms   213.121.98.144
  5     14 ms      10 ms      10 ms   87.237.20.142
  6     10 ms      10 ms      12 ms   72.14.242.70
  7     10 ms      10 ms       9 ms   74.125.242.65
  8     10 ms      10 ms      10 ms   142.251.52.143
  9     10 ms       9 ms      10 ms   dns.google [8.8.8.8]
```

Screenshot courtesy of Microsoft.

- tracert command
 - RTT per hop
- pathping command
 - Can show latency and packet loss per hop

Troubleshoot Name Resolution (Slide 1 of 2)



Connection via FQDN vs. connection with IP address directly



nslookup command

-type=<resourcerecordtype> where
<resourcerecordtype> = A, AAAA,
MX, NS, PTR, CNAME etc.

Troubleshoot Name Resolution (Slide 2 of 2)

```
C:\Users\Admin>nslookup -type=mx comptia.org 8.8.8.8
Server:  dns.google
Address:  8.8.8.8

Non-authoritative answer:
comptia.org      MX preference = 10, mail exchanger = comptia-org.mail.protection.outlook.com

C:\Users\Admin>nslookup -type=ns comptia.org 8.8.8.8
Server:  dns.google
Address:  8.8.8.8

Non-authoritative answer:
comptia.org      nameserver = ns2.comptia.org
comptia.org      nameserver = ns1.comptia.org

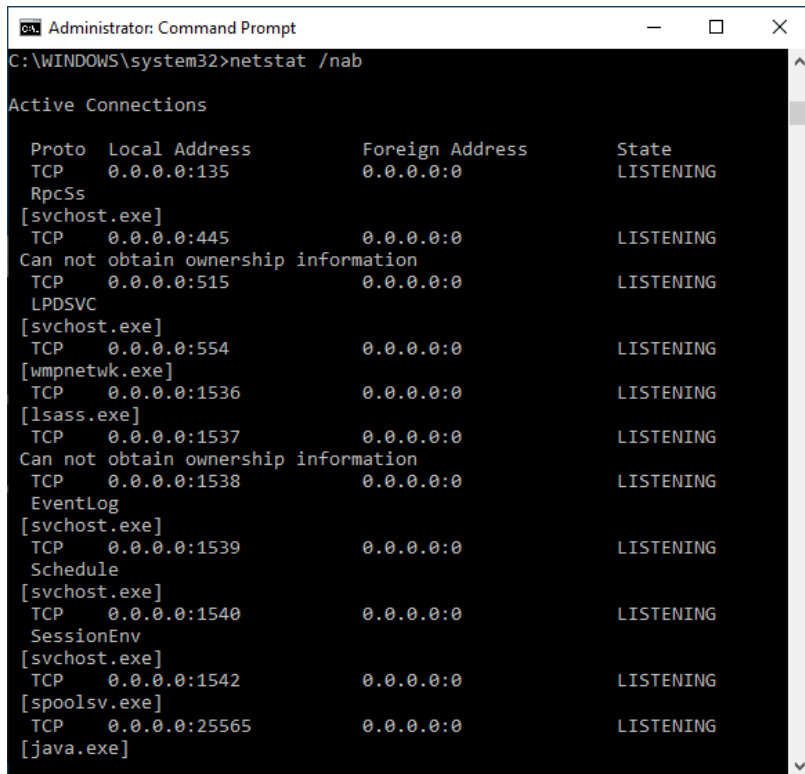
C:\Users\Admin>nslookup -type=mx comptia.org ns1.comptia.org
Server:  UnKnown
Address:  209.117.62.56

comptia.org      MX preference = 10, mail exchanger = comptia-org.mail.protection.outlook.com

C:\Users\Admin>_
```

Screenshot courtesy of Microsoft.

Troubleshoot Network Ports

A screenshot of a Windows Command Prompt window titled "Administrator: Command Prompt". The command prompt shows the command "C:\WINDOWS\system32>netstat /nab" and its output. The output is titled "Active Connections" and lists several listening TCP connections. Each entry includes the protocol (TCP), the local address (IP and port), the foreign address (IP and port), and the state (LISTENING). The processes associated with each connection are listed in brackets, such as [svchost.exe], [wmpnetwk.exe], [lsass.exe], [EventLog], [Schedule], [SessionEnv], [spoolsv.exe], and [java.exe]. Some entries are preceded by the text "Can not obtain ownership information".

```
Administrator: Command Prompt
C:\WINDOWS\system32>netstat /nab

Active Connections

 Proto Local Address           Foreign Address         State
 TCP   0.0.0.0:135               0.0.0.0:0               LISTENING
      RpcSs
 [svchost.exe]
 TCP   0.0.0.0:445               0.0.0.0:0               LISTENING
 Can not obtain ownership information
 TCP   0.0.0.0:515               0.0.0.0:0               LISTENING
      LPDSVC
 [svchost.exe]
 TCP   0.0.0.0:554               0.0.0.0:0               LISTENING
 [wmpnetwk.exe]
 TCP   0.0.0.0:1536              0.0.0.0:0               LISTENING
 [lsass.exe]
 TCP   0.0.0.0:1537              0.0.0.0:0               LISTENING
 Can not obtain ownership information
 TCP   0.0.0.0:1538              0.0.0.0:0               LISTENING
      EventLog
 [svchost.exe]
 TCP   0.0.0.0:1539              0.0.0.0:0               LISTENING
      Schedule
 [svchost.exe]
 TCP   0.0.0.0:1540              0.0.0.0:0               LISTENING
      SessionEnv
 [svchost.exe]
 TCP   0.0.0.0:1542              0.0.0.0:0               LISTENING
 [spoolsv.exe]
 TCP   0.0.0.0:25565             0.0.0.0:0               LISTENING
 [java.exe]
```

Screenshot courtesy of Microsoft.

- netstat command
 - -a include UDP connections
 - -b show process that opened port
 - -n numerical format
 - -e report ethernet statistics
 - -s report protocol statistics

Lab Activity

- Lab: Assisted Troubleshooting
(available in CertMaster Learn and CertMaster Perform)
- Lab: Troubleshoot Physical Connectivity
(available in CertMaster Learn and CertMaster Perform)
- Lab: Connectivity
(available in CertMaster Learn and CertMaster Perform)
- Lab: Services/Name Resolution
(available in CertMaster Learn and CertMaster Perform)

Lesson Review: Troubleshoot Windows Networking

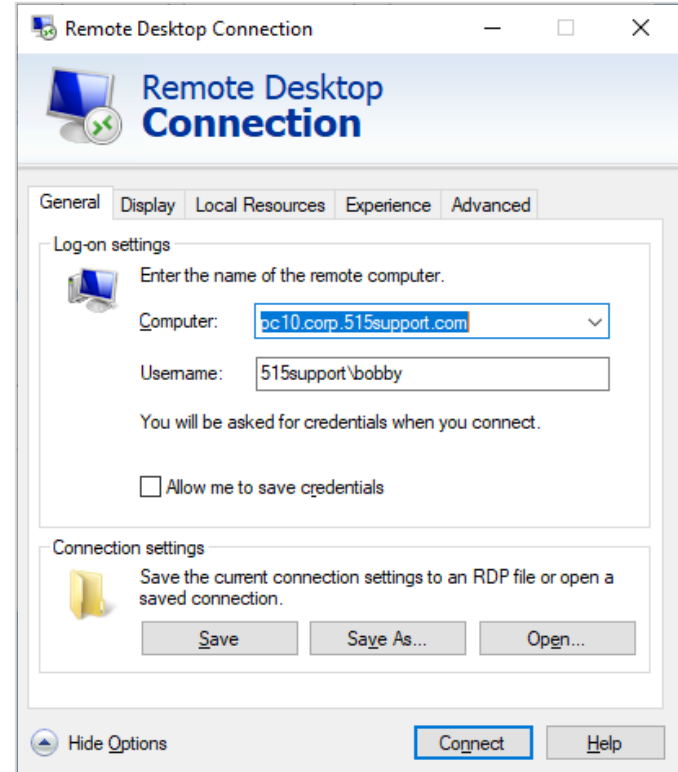
- Troubleshoot IP Configuration
- Troubleshoot Local Network Connectivity
- Troubleshoot Remote Network Connectivity
- Troubleshoot Name Resolution
- Troubleshoot Network Ports

Lesson 14.2

Remote Access Technologies

Remote Desktop Tools

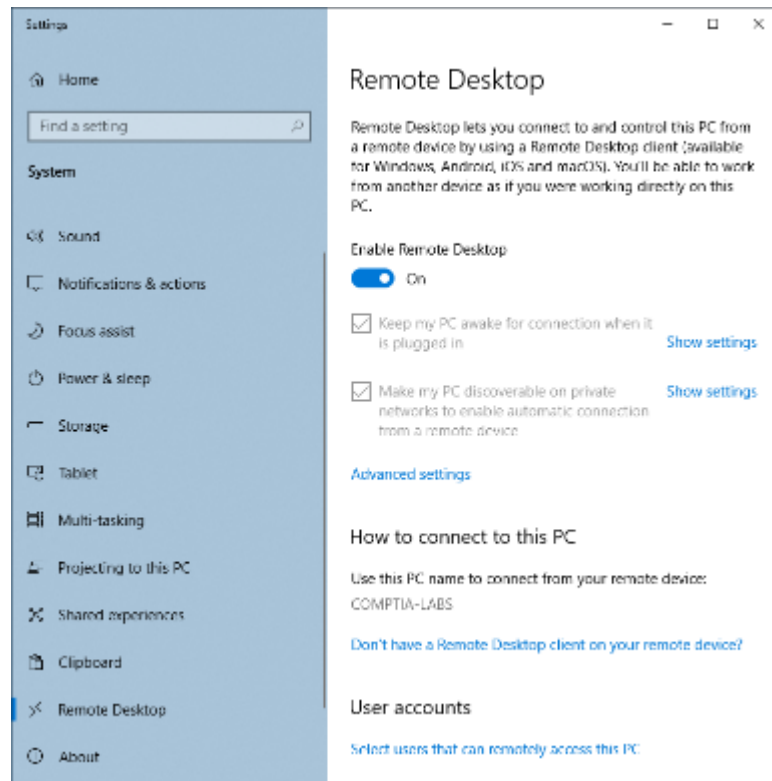
- Remote Desktop Protocol (RDP)
 - Remote Desktop Connection application (mstsc.exe)
 - TCP port 3389
 - Encrypted authentication and session
 - IP or FQDN of remote system needed
 - Credentials to log in
- Virtual Network Computing (VNC)
 - TCP port 5900



Screenshot courtesy of Microsoft.

RDP Server and Security Settings

- Settings > Remote Desktop
- Client available in all editions
- RDP Server only available in Pro and Enterprise editions

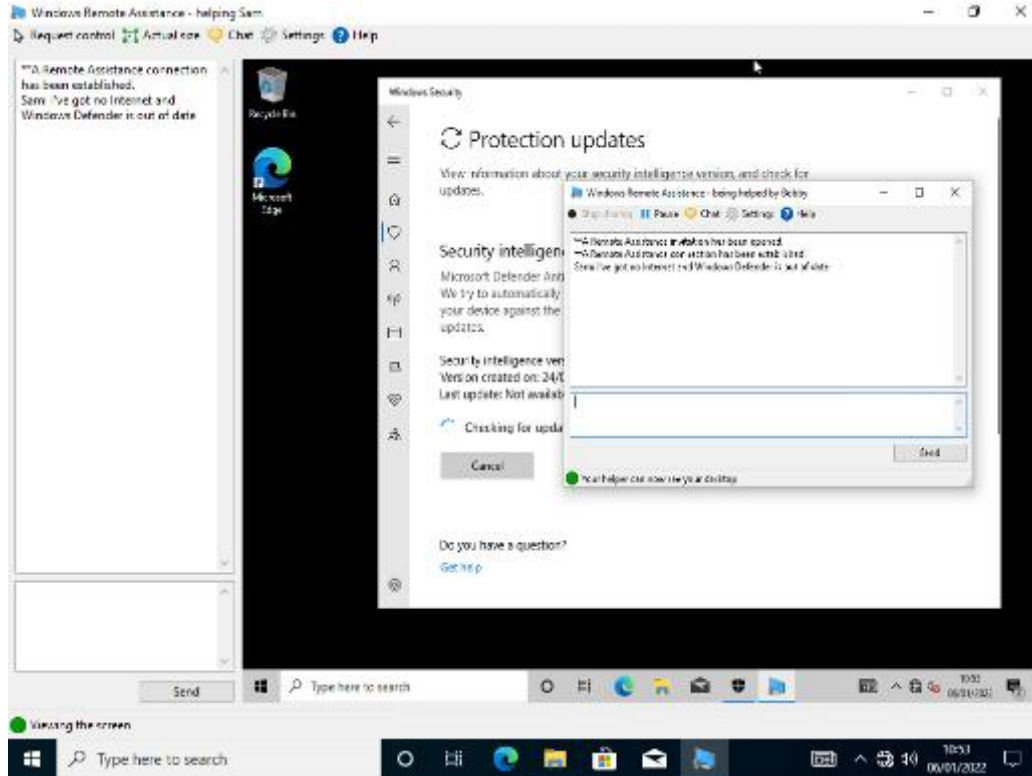


Screenshot courtesy of Microsoft.

Microsoft Remote Assistance (Slide 1 of 2)

- Initiated by invite only
- Includes chat feature and remote-control capability
- Ephemeral port use (49152-65535)
- Quick Assist
 - Facilitates Remote Assistance without reconfiguring firewall using TCP port 443 (HTTPS)

Microsoft Remote Assistance (Slide 2 of 2)



Screenshot courtesy of Microsoft.

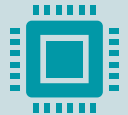
WinRM



Uses SOAP-based communication



Allows access and exchange of management information across network



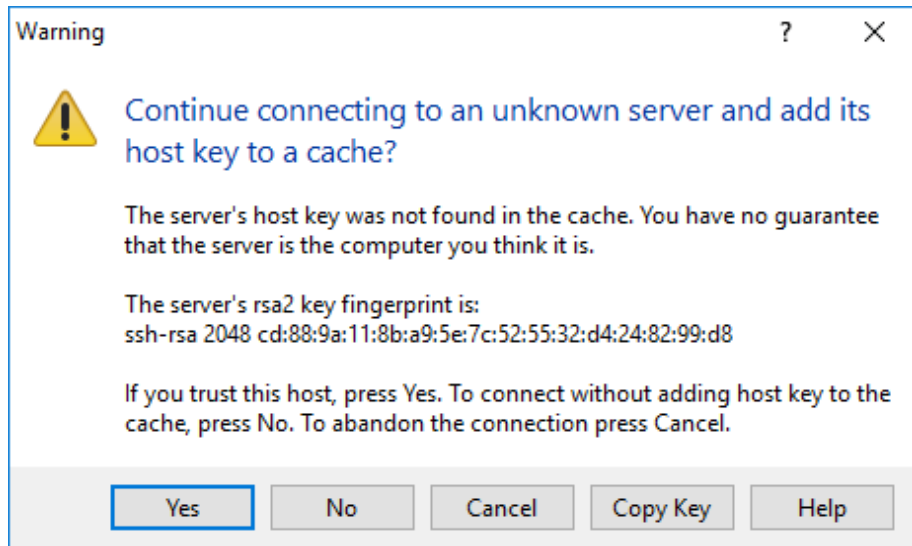
Interoperability between different hardware and software platforms using HTTP/HTTPS



Can be easily used in PowerShell scripts

Secure Shell

- TCP port 22
- Encrypted
- Password or key-based authentication methods



Screenshot courtesy of Microsoft.

Desktop Management and Remote Monitoring Tools

Remote monitoring and management (RMM)

- Primarily an MSP tool for remote support

Desktop management and MDM

- Management of device belonging to a single organization

Endpoint Detection and Response (EDR) capability

- Security tool for scans and reporting of issues with pre-programmed responses

Simple Protocol for Independent Computing Environments (SPICE)

- Remote display to monitor and interact with a VM
- Authentication support for Kerberos and others
- TLS encryption of communication

Other Remote Access Tools

- Screen sharing
 - TeamViewer or LogMeIn
- Video conferencing tools
 - Microsoft Teams or Zoom
 - Screen share capabilities
- File transfer software
 - AirDrop for Apple
 - Nearby Sharing for Microsoft
 - Nearby Share for Android
- VPN
 - Encrypted tunnel to connect to internal systems through the VPN gateway/server

Lab Activity

- Lab: Configure Remote Services
(available in CertMaster Learn and CertMaster Perform)
- Applied Live Lab: Provide Remote Support
(available in CertMaster Perform and CertMaster Labs)

Lesson Review: Remote Access Technologies

- Remote Desktop Tools
- RDP Server and Security Settings
- Microsoft Remote Assistance
- WinRM
- Secure Shell
- Desktop Management and Remote Monitoring Tools
- Simple Protocol for Independent Computing Environments (SPICE)
- Other Remote Access Tools

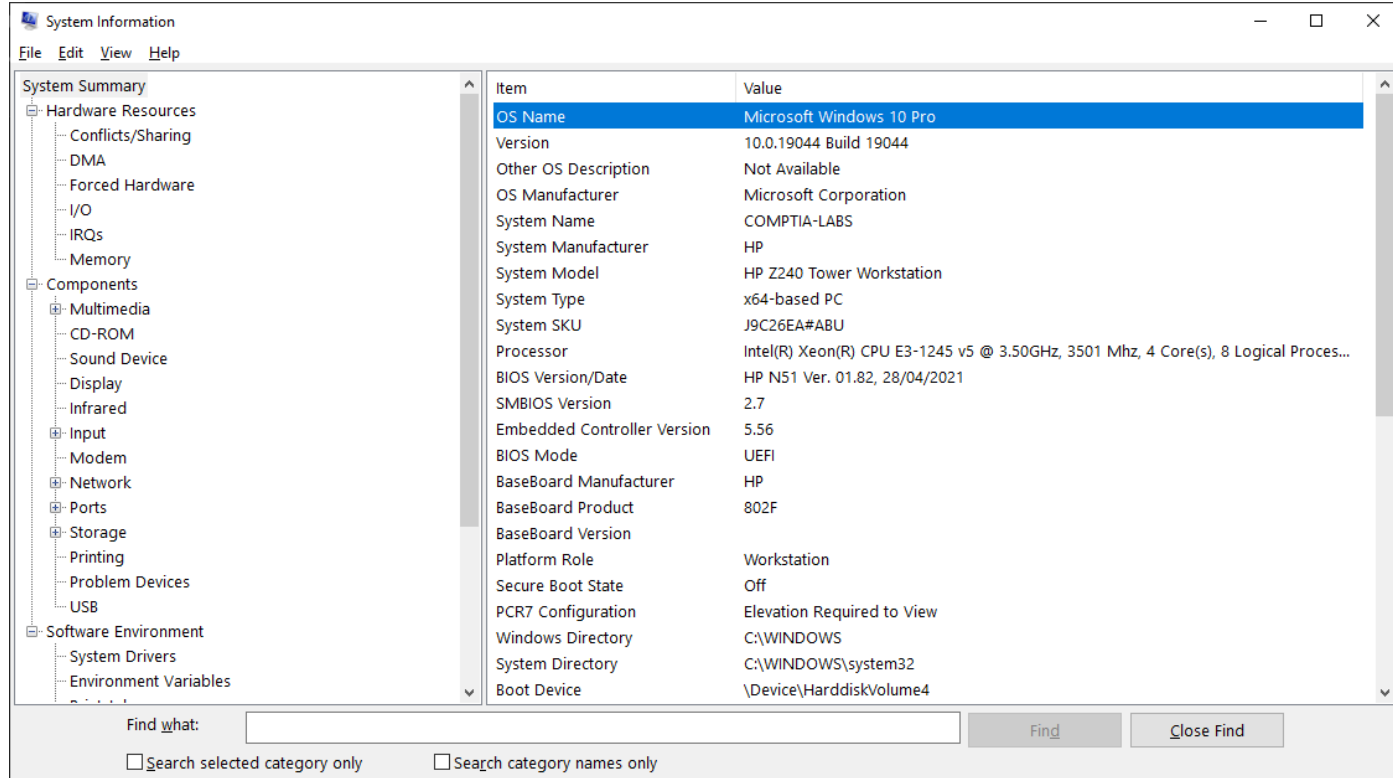
Lesson 14.3

Performance and Troubleshooting Tools

System Information (Slide 1 of 2)

- msinfo32.exe
- System resources
- Firmware version
- OS version
- Hardware
- Environment variables
- Network status

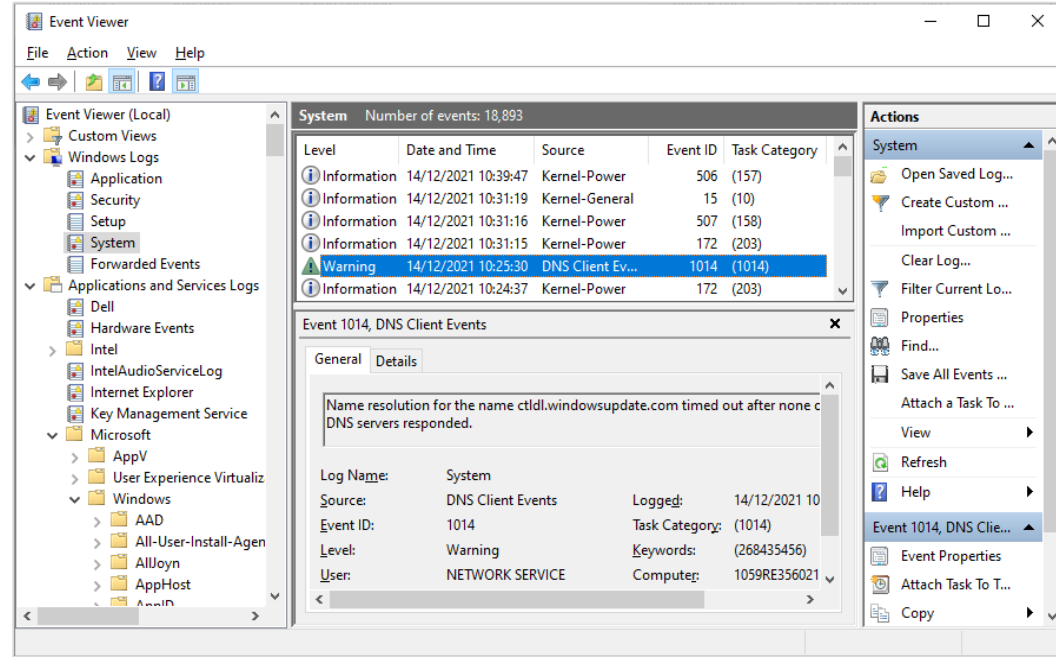
System Information (Slide 2 of 2)



Screenshot courtesy of Microsoft.

Event Viewer

- Eventvwr.msc
- Windows Log viewing
 - Application
 - Security
 - Setup
 - System
 - Forwarded Events
- Severity Levels
 - Information, Warning, Error, Critical



Screenshot courtesy of Microsoft.

Task Manager Process Monitoring (Slide 1 of 2)

- CTRL+SHIFT+ESC
- CTRL+ALT+DEL then select Task Manager
- Right-click Start button or the taskbar
- Processes
- Details
- Set Priority

Task Manager Process Monitoring (Slide 2 of 2)

The screenshot displays the Windows Task Manager application window. The 'Performance' tab is selected, showing a summary of system resources: CPU at 3%, Memory at 29%, Disk at 0%, Network at 0%, GPU at 0%, and Power usage at 'Very low'. The 'Processes' tab is also visible, showing a list of running applications and background processes. A context menu is open over the 'Notepad' process, offering options such as 'Expand', 'End task', 'Resource values', 'Provide feedback', 'Debug', 'Create dump file', 'Go to details', 'Open file location', 'Search online', and 'Properties'. The 'End task' option is highlighted. The 'Background processes' section is also visible, listing various system services.

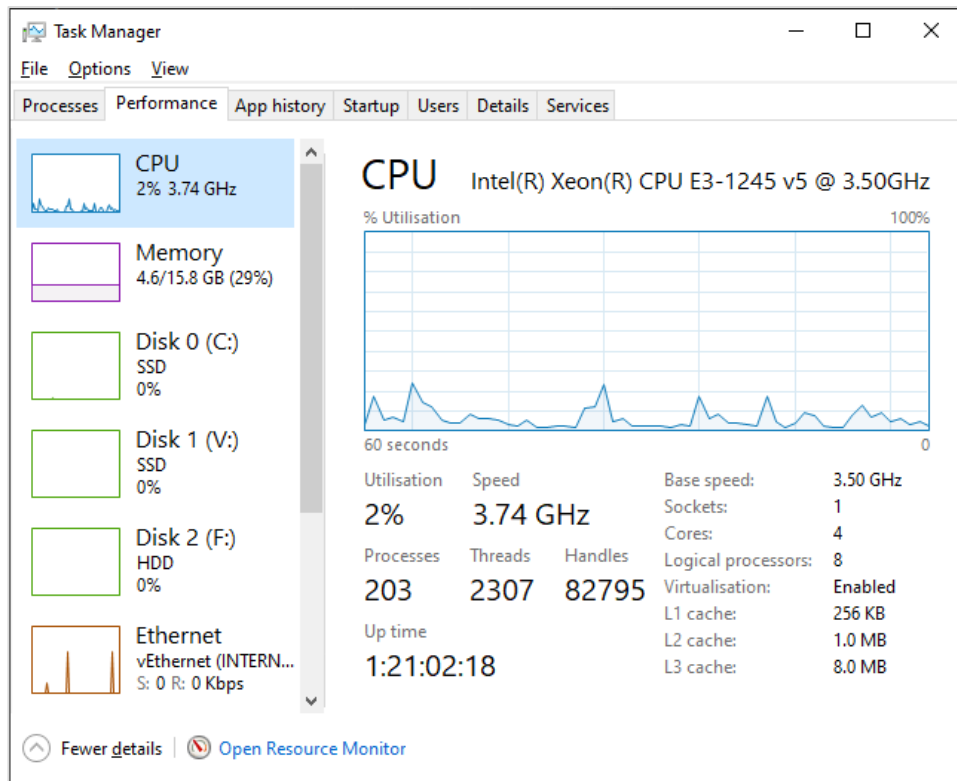
Name	Status	3% CPU	29% Memory	0% Disk	0% Network	0% GPU	GPU engine	Power usage
Apps (4)								
> Mail	🟢	0%	0 MB	0 MB/s	0 Mbps	0%	GPU 1 - 3D	Very low
> Notepad	🟢	0%	0 MB	0 MB/s	0 Mbps	0%	GPU 1 - 3D	Very low
> Settings	🟢	0%	0 MB	0 MB/s	0 Mbps	0%	GPU 1 - 3D	Very low
> Task Manager	🟢	0%	0 MB	0 MB/s	0 Mbps	0%		Very low
Background processes (4)								
> 64-bit Synaptics Pointing Device	🟢	0%	0 MB	0 MB/s	0 Mbps	0%		Very low
> Adobe Acrobat Update Service	🟢	0%	0 MB	0 MB/s	0 Mbps	0%		Very low
> Antimalware Service Executable	🟢	0%	0 MB	0 MB/s	0 Mbps	0%		Very low
> Application Frame Host	🟢	0%	0 MB	0 MB/s	0 Mbps	0%		Very low
> COM Surrogate	🟢	0%	0 MB	0 MB/s	0 Mbps	0%		Very low

Screenshot courtesy of Microsoft.

Task Manager Performance Monitoring (Slide 1 of 2)

- App history
 - Windows Store apps
- Disk monitoring
- Network monitoring

Task Manager Performance Monitoring (Slide 2 of 2)



Screenshot courtesy of Microsoft.

CPU and GPU Monitoring

- CPU
 - Cores and logical processors (HyperThreading)
 - Virtualization status
- GPU
 - Dedicated graphics adapter monitoring
 - Graphics memory monitor

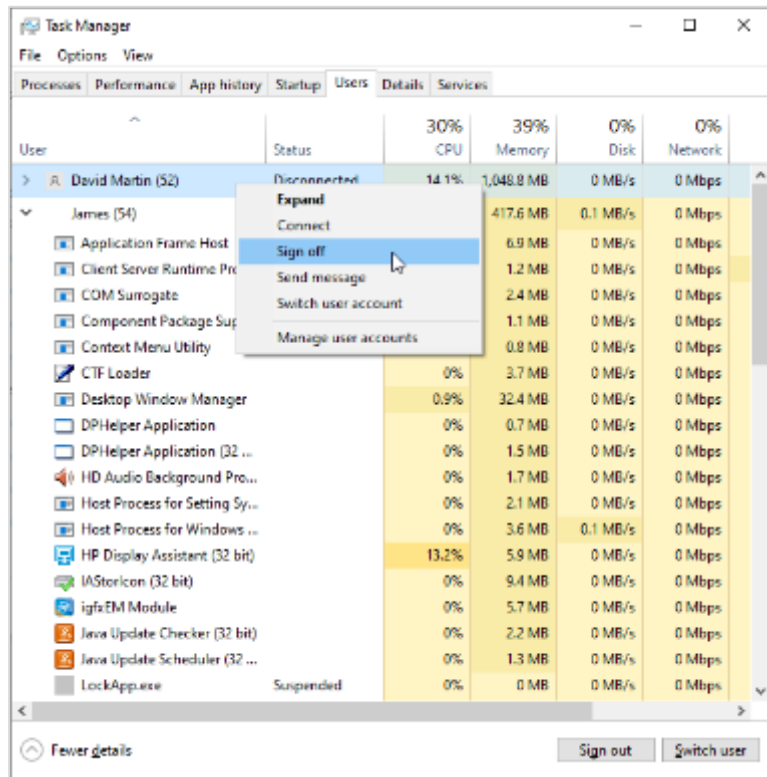
Memory Monitoring

- In use
- Committed
- Cached
- Paged pool/non-paged pool
- High utilization can be normal to maximize performance

Disk and Network Monitoring

- Disk
 - Type, capacity, and statistics
- Network
 - IP and MAC information along with statistics
 - SSID, signal strength, and standard for Wi-Fi connections

Task Manager User Monitoring



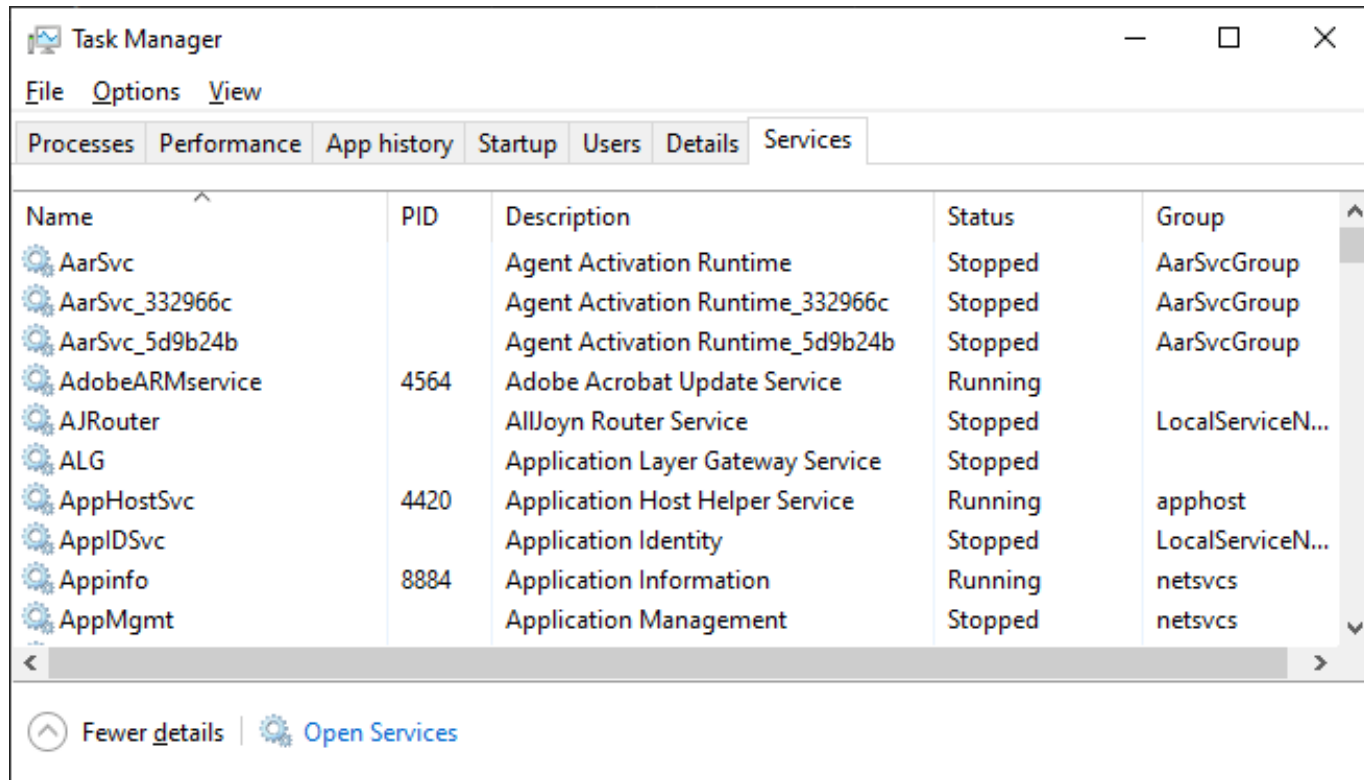
Screenshot courtesy of Microsoft.

- View logged-in users
- Send messages
- Sign out users
- View running processes for users
- Resource monitoring on per-user basis

Startup Processes and Services Console (Slide 1 of 2)

- Startup allows disabling programs from starting when Windows launches
- Services allows for monitoring of background services
 - Disable non-essential services for better performance and security

Startup Processes and Services Console (slide 2 of 2)



Screenshot courtesy of Microsoft.

Resource Monitor and Performance Monitor (Slide 1 of 2)

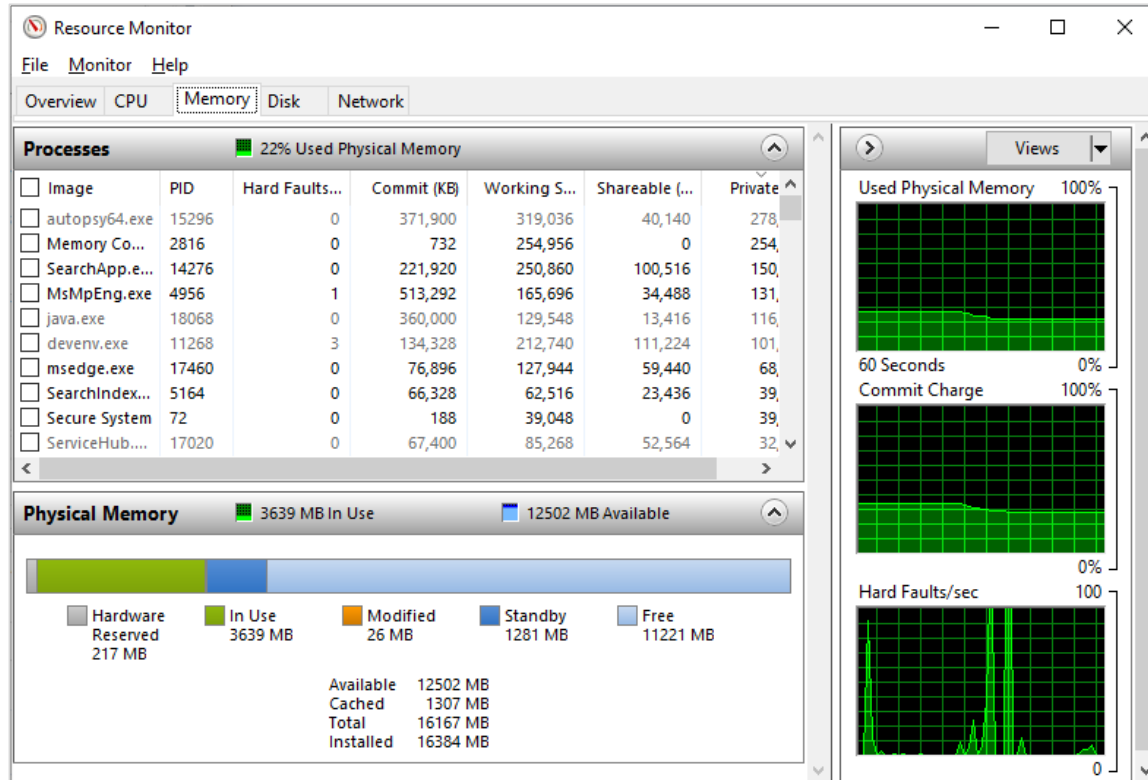
Resource Monitor (resmon.exe)

- Advanced monitoring of resources beyond Task Manager's capabilities

Performance Monitor (perfmon.exe)

- Real-time statistic charts and logs for long-term analysis
- Counter logs
- Trace logs

Resource Monitor and Performance Monitor (Slide 2 of 2)

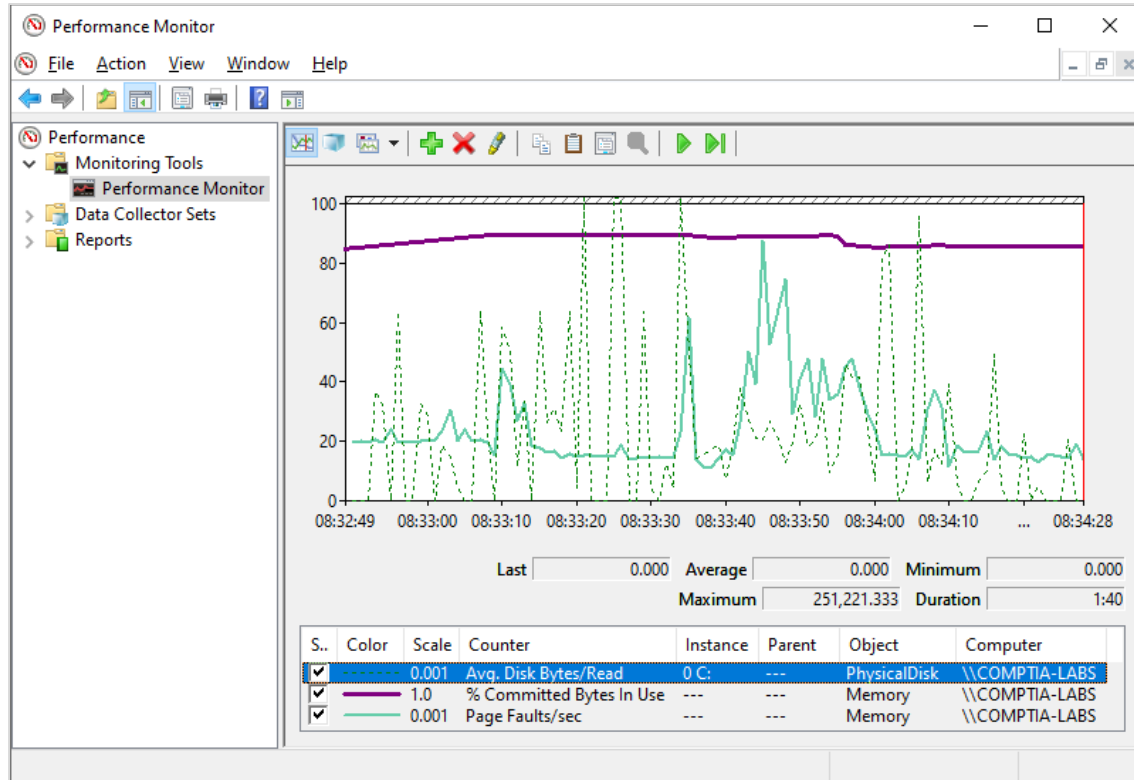


Screenshot courtesy of Microsoft.

Performance Counters (Slide 1 of 2)

- Specific monitoring metrics to be used in Performance Monitor utility
- Examples
 - % processor time
 - Average disk queue length
 - Memory pages/sec
 - Paging file % usage

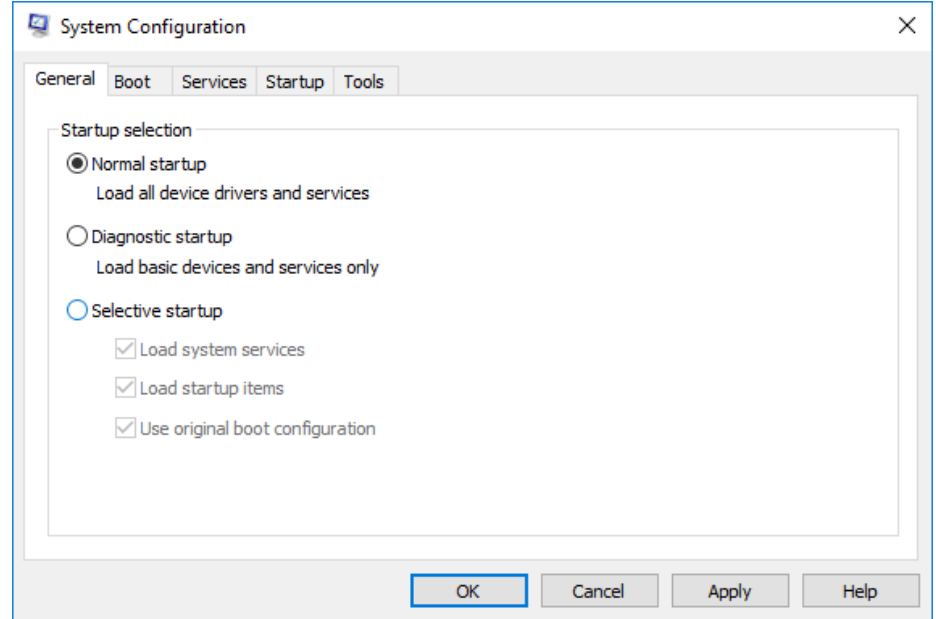
Performance Counters (Slide 2 of 2)



Screenshot courtesy of Microsoft.

System Configuration Utility

- General
- Boot
- Services
- Tools



Screenshot courtesy of Microsoft.

Lab Activity

- Lab: Manage Applications
(available in CertMaster Learn and CertMaster Perform)

Lesson Review: Performance and Troubleshooting Tools

- System Information
- Event Viewer
- Task Manager Process Monitoring
- Task Manager Performance Monitoring
- CPU and GPU Monitoring
- Memory Monitoring
- Disk and Network Monitoring
- Task Manager User Monitoring
- Startup Processes and Services Console
- Resource Monitor and Performance Monitor
- Performance Counters
- System Configuration Utility

Lesson 14.4

Troubleshoot Windows OS Problems

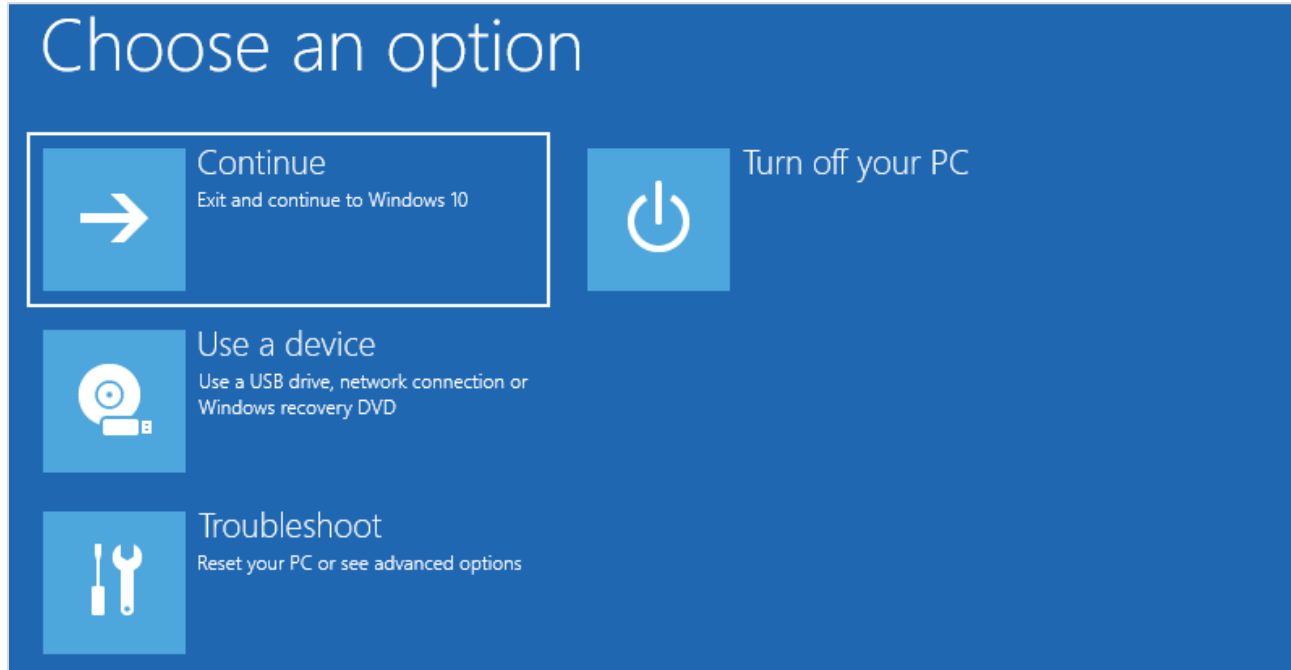
Boot Process

- BIOS
 - POST
 - Boot loader- BOOTMGR.EXE
 - WINLOAD.EXE
 - Kernel
 - HAL.DLL
 - Drivers
- UEFI
 - POST
 - GPT read
 - BCD and BOOTMGFW.EFI
 - Kernel
 - HAL.DLL
 - Drivers

Boot Recovery Tools (Slide 1 of 4)

- Advanced Boot Options
 - Interrupt boot process
 - Hold shift while selecting restart from power menu in Windows
- Troubleshoot
- WinRE
- Start-up repair
- Advanced options

Boot Recovery Tools (Slide 2 of 4)



Screenshot courtesy of Microsoft.

Boot Recovery Tools (slide 3 of 4)

Startup Settings

Press a number to choose from the options below:

Use number keys or functions keys F1-F9.

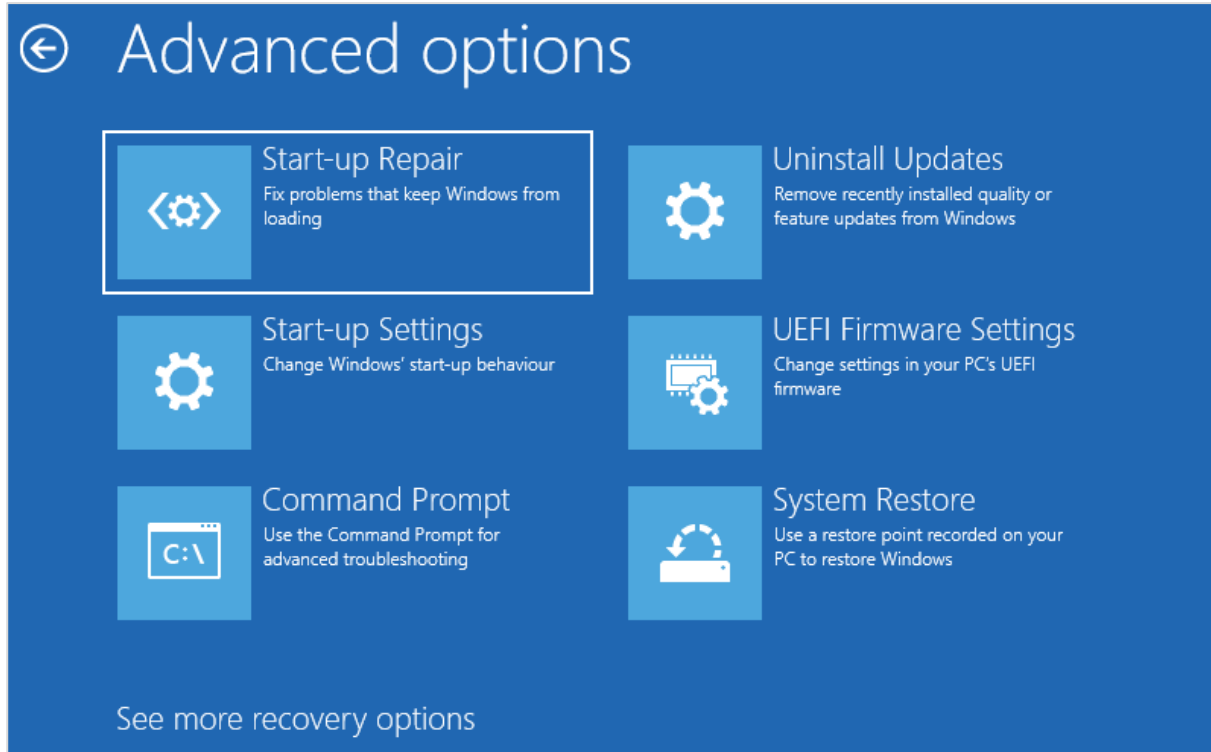
- 1) Enable debugging
- 2) Enable boot logging
- 3) Enable low-resolution video
- 4) Enable Safe Mode
- 5) Enable Safe Mode with Networking
- 6) Enable Safe Mode with Command Prompt
- 7) Disable driver signature enforcement
- 8) Disable early launch anti-malware protection
- 9) Disable automatic restart after failure

Press F10 for more options

Press Enter to return to your operating system

Screenshot courtesy of Microsoft.

Boot Recovery Tools (slide 4 of 4)

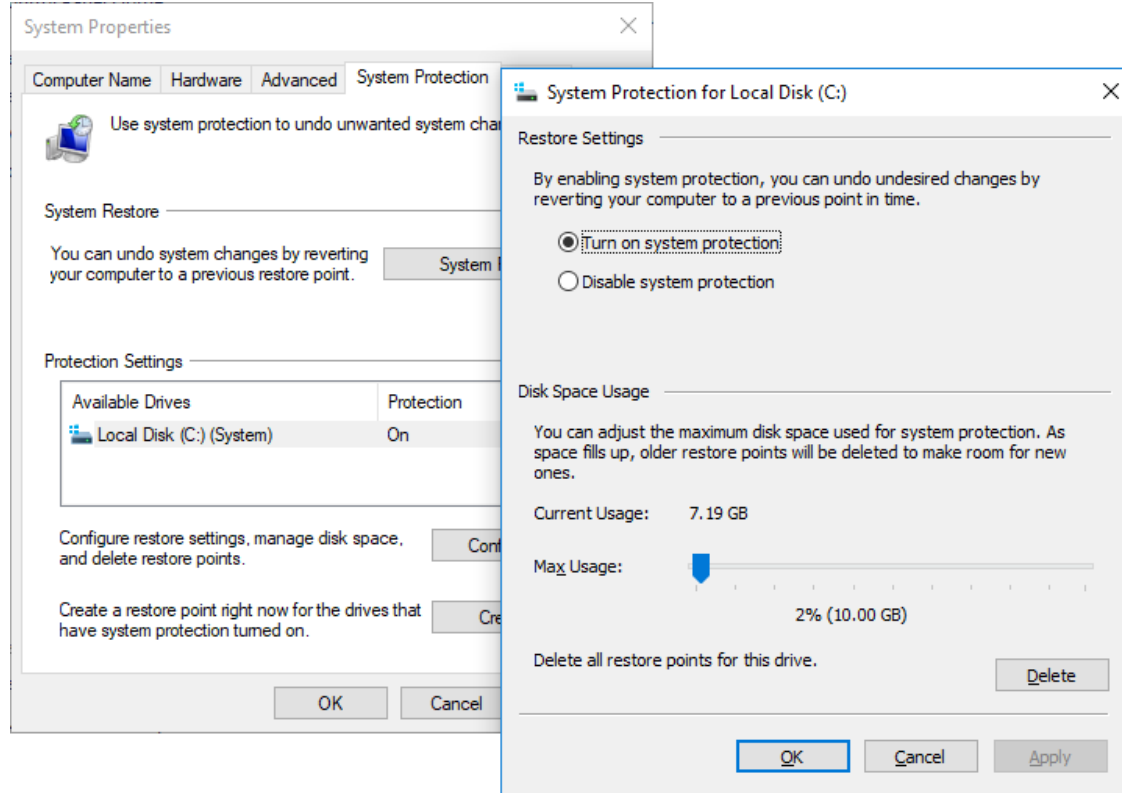


Screenshot courtesy of Microsoft.

System Restore (Slide 1 of 3)

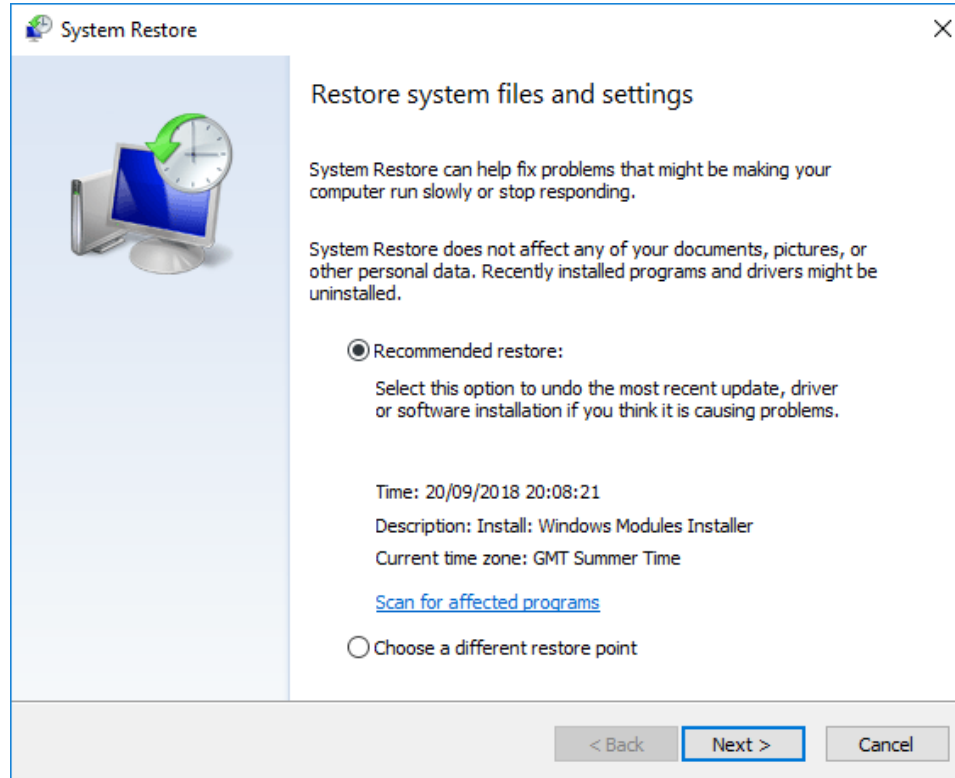
- Configure system protection
 - Restore point
 - Disk space usage
- Using system restore

System Restore (Slide 2 of 3)



Screenshot courtesy of Microsoft.

System Restore (Slide 3 of 3)



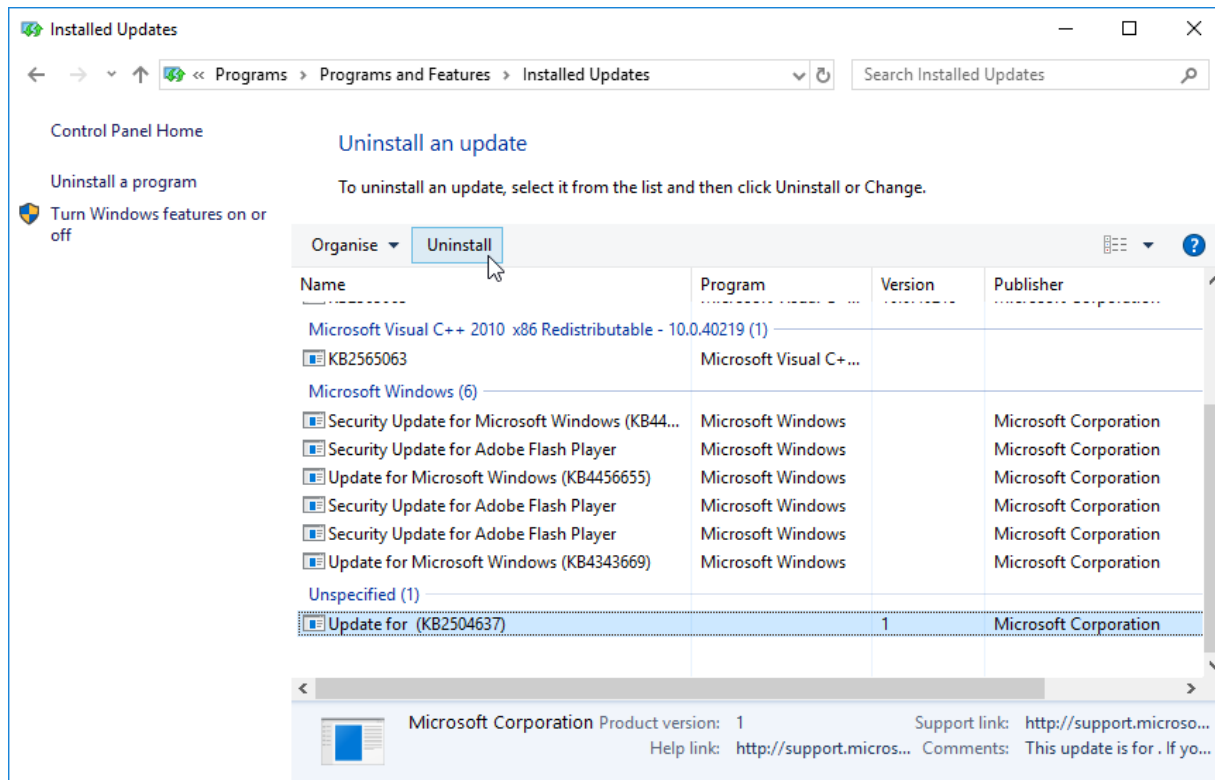
Screenshot courtesy of Microsoft.

Update and Driver Roll Back (Slide 1 of 3)

Programs and
Features applet > view
installed updates >
select update and
then uninstall

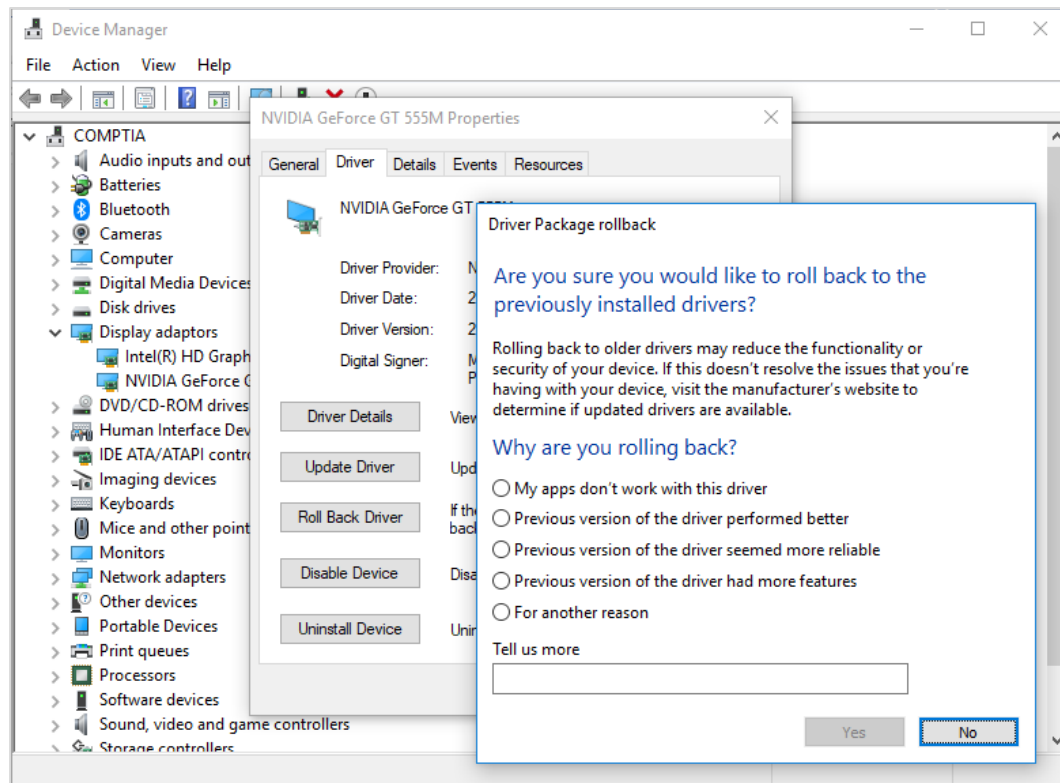
Device Manager >
Properties > Driver >
Roll Back Driver

Update and Driver Roll Back (Slide 2 of 3)



Screenshot courtesy of Microsoft.

Update and Driver Roll Back (Slide 3 of 3)



Screenshot courtesy of Microsoft.

System Repair, Reinstall, and Reimage (Slide 1 of 2)

- Recovery image
 - Backup of the system used to restore system config and files
- Reinstall Windows
 - Reset this PC option in WinRE
 - Keep data files or remove everything

System Repair, Reinstall, and Reimage (Slide 2 of 2)

⬅ Reset this PC

Choose an option



Keep my files

Removes apps and settings, but keeps your personal files.



Remove everything

Removes all of your personal files, apps and settings.

Screenshot courtesy of Microsoft.

Troubleshoot Boot Issues

- Failure to boot
 - Verify no floppy disk is installed or USB is plugged in still and set to boot
- No OS Found
 - Use chkdsk and bootrec commands from the command prompt
 - diskpart could also be used to verify system partition is set to active
- GUI Fails to Load
 - Verify monitor turned on and connected to correct video port
 - Verify input selection
 - Boot to Safe Mode

01

Display highly detailed status messages

02

Verify drivers and services set to load immediately after startup

03

Rebuild the local profile, if necessary

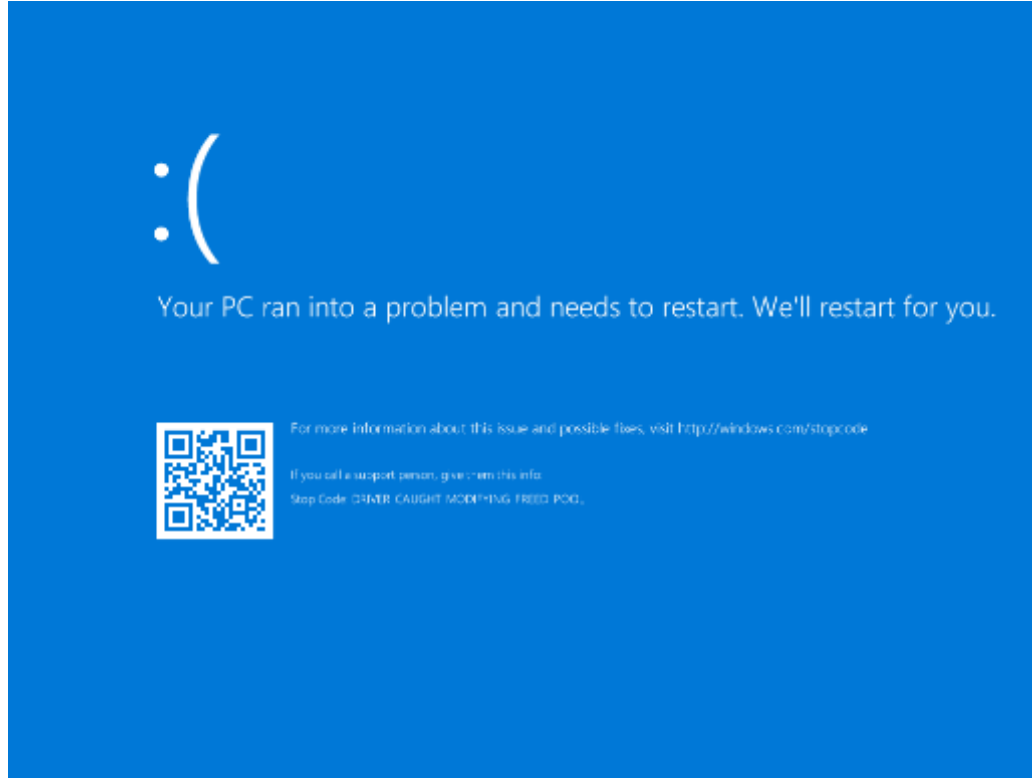
Troubleshoot Performance Issues

- Degraded Performance
 - Task Manager, PerfMon, and ResMon can be used to verify utilization
 - Reboot the system
 - Update OS, applications, and drivers as they may improve performance by fixing known issues
 - Scan for malware and verify antivirus/antimalware updates
 - Power management issues

Troubleshoot System Fault Issues (Slide 1 of 2)

- Blue Screen of Death (BSOD)
 - Research indicated problem from details on screen or using QR code
- System instability
- Frequent unplanned shutdowns
 - Overheating or corruption of system files
- USB Issues
 - Controller resource issues

Troubleshoot System Fault Issues (Slide 2 of 2)



Screenshot courtesy of Microsoft.

Troubleshoot Application and Service Fault Issues

- Application crashes
 - Update
 - If no change, uninstall and reinstall
- Services not starting
 - Verify service is set to automatically start, if necessary
- Time drift
 - Replace RTC battery (CR 2032)
 - Synchronize clock with known good source (NTP server)

Lab Activity

- Lab: Boot into the Windows Recovery Environment
(available in CertMaster Learn and CertMaster Perform)
- Lab: Create a Restore Point
(available in CertMaster Learn and CertMaster Perform)
- Lab: Configure the Boot Order
(available in CertMaster Learn and CertMaster Perform)
- Applied Live Lab: Support Windows OS
(available in CertMaster Perform and CertMaster Labs)
- Challenge Live Lab: Resolve Troubleshooting Tickets
(available in CertMaster Perform and CertMaster Labs)
- Challenge Live Lab: Resolve Network Troubleshooting Tickets
(available in CertMaster Perform and CertMaster Labs)

Lesson Review: Troubleshoot Windows OS Problems

- Boot Process
- Boot Recovery Tools
- System Restore
- Update and Driver Roll Back
- System Repair, Reinstall, and Reimage
- Troubleshoot Boot Issues
- Troubleshoot Profile Issues
- Troubleshoot Performance Issues
- Troubleshoot System Fault Issues
- Troubleshoot Application and Service Fault Issues